MDL A4:

Part A:

Given,

$P(\text{heads}) = P$

$P(\text{tails}) = 1 - P$

Probability that odd no. of heads after n independent tosses $= q_n$

(i) Derivation:

Case-I: There will be odd no. of heads after "n-1" tosses, and then n^{th} toss should or must be Tails.

(or)

Case-II: There will be even no. of heads after "n-1" tosses and then n^{th} toss should or must be heads.

So, from above definition of q_n,

The total probability of getting odd no. of heads after "n" tosses

= probability of Case-I + Case-II.

hence,

$$q_n = q_{n-1} \cdot (1-P) + (1-q_{n-1}) \cdot P.$$

q_{n-1} = odd no. of heads after "n-1" tosses

$1-q_{n-1}$ = even no. of heads after "n-1" tosses.

$$q_n = q_{n-1} - P \cdot q_{n-1} + P - P \cdot q_{n-1}.$$

$$\boxed{q_n = q_{n-1}(1-2P) + P}$$

($q_0 = 0$, $q_1 = P$)
↳ Base cases.

This is requested recursive relation.

(ii) from above equation

$$q_n = q_{n-1}(1-2P) + P$$

To solve this recursion, we use telescoping + iteration

$$q_n = \cancel{q_{n-1}(1-2P)} + P$$

$$\cancel{q_{n-1}(1-2P)} = \cancel{q_{n-2}(1-2P)^2} + P(1-2P)$$

$$\vdots$$

$$\cancel{q_{n-r}(1-2P)^r} = \cancel{q_{n-r-1}(1-2P)^{r+1}} + P(1-2P)^r$$

$$\vdots$$

$$\cancel{q_1(1-P)^{n-1}} = \cancel{q_0}^{\,0}(1-2P)^n + P(1-2P)^{n-1}$$

By adding all the terms,

r^{th} term is :

$$q_{n-r}(1-2p)^r = q_{n-r-1}\cdot(1-2p)^{r+1} + p(1-2p)^r$$

Finally we get,

$$\boxed{q_n = p(1) + p(1-2p) + p(1-2p)^2 + \cdots + p(1-2p)^{n-1} + q_0}$$

From base case,

$q_0 = 0$ (as without toss odd's heads probability is zero).

$$q_n = p\left(1 + (1-2p) + (1-2p)^2 + \cdots + (1-2p)^{n-1}\right)$$

$$q_n = p\left(\frac{((1-2p)^n - 1)}{(1-2p) - 1}\right)$$

$$\sum_{n=0}^{n=n-1} r^{n-1} = \frac{r^n - 1}{r-1}$$

$$q_n = p\left(\frac{(1-2p)^n - 1}{-2p}\right) \Rightarrow \frac{\cancel{p}\left(1-(1-2p)^n\right)}{2\cancel{p}}$$

$$\boxed{q_n = \frac{1-(1-2p)^n}{2}}$$

Q 3.1) Let G_1 be the event that generator 1 fails, &
Let G_2 be the event that generator 2 fails.

Each has a 3% chance of failing in 12 hours.

$\Rightarrow P(G_1) = 0.03$, and $P(G_2) = 0.03$.

The generators operate independently.

The hospital will only lose power if both backup generators fail during 12-hour period.

$\therefore P(\text{Losing Power}) = P(G_1 \cap G_2)$.

$P(G_1 \cap G_2) = P(G_1) \cdot P(G_2)$. (Independent events)

$P(G_1 \cap G_2) = (0.03) \cdot (0.03)$

$= 0.0009$

$\therefore P(\text{Losing Power}) = 0.0009 = (0.09\%)$

Q. 3.2).

Let $P(\text{Same})$ be the probability that at least two people share the same birthday. Using the rule of complements, it is easier to calculate the probability that nobody shares a birthday. (i.e, all ~~have~~ 25 have unique birthday), and subtract it from 1: $P(\text{Same}) = 1 - P(\text{all unique})$.

To find $P(\text{all unique})$:

- The first person can have any of the 365 days: $\frac{365}{365}$.
- The second person can have a different day from first $\frac{364}{365}$
- The 3rd person can have a different day from 1st & 2nd: $\frac{363}{365}$

⋮

- The 25th person can have a different day from first 24: $\frac{341}{365}$

$$\therefore P(\text{all unique}) = \frac{365 \times 364 \times 363 \times \cdots \times 341}{365^{25}}$$

$$= \frac{(365)!}{(365-25)! \cdot (365^{25})}$$

$$= \frac{{}^{365}C_{25}}{365^{25}}$$

$\therefore P(\text{all unique}) = \dfrac{{}^{365}C_{25}}{365^{25}}$

$\approx 0.431300\ldots\ldots$

$\therefore P(\text{Same}) = 1 - P(\text{all unique})$

$= 1 - 0.431300\ldots$

≈ 0.5687

$\therefore$ The probability is approximately 0.5687 or (56.87%)